

- c. Inference is constrained by regularities in the conversational domain that give rise to shared assumptions.

In other words, communication involves a fundamental layer of inference to the best explanation (i.e. non-demonstrative inference). Such inference involves uncertainty, as it can not be guaranteed that the listener will find the best explanation for the speaker's behaviour. However, inference in communication takes place against the background of regularities characteristic of rational action which are implicitly shared by speakers and listeners and make their actions transparent to each other.

Grice sought to explain inference in verbal communication by identifying a set of maxims that rational communicators must be expected to work under. Rational Speech-Act theory (RSA)³ formalizes this inferential account of communication as a series of interacting probabilistic reasoning processes. Informally, we can outline this intertwined reasoning process as it applies to our example as in (16)⁴:

- (16) Mary's and Peter's reasoning based on the assumption that speakers and listeners are rational communicators and follow the maxims as much as possible:

MARY: ["pragmatic speaker" reasoning about a "pragmatic listener"]

- (a) I want to make it manifest to Peter that I don't want to drink coffee at 6 pm.
- (b) I also want to make it manifest to Peter that I do want something to drink and that I do want to be sociable at the party.
- (c) I need to choose the utterance that maximizes the probability that Peter, assuming that I follow the maxims, can interpret in a way that satisfies the maxims and conveys propositions that express my intentions/goals.

PETER: ["pragmatic listener" reasoning about a "pragmatic speaker" and how such a speaker differs from a "literal speaker" who in turn reasons about a "literal listener"]

- (a) I expect Mary to confirm or reject that Mary wants to drink a cup of coffee at 6 pm.
- (b) I expect Mary to be sociable even when declining the offer.
- (c) Mary is a rational communicator and follows the maxims.
- (d) Mary's utterance, considering only what is said in it, does not satisfy my expectation that she confirms or rejects the idea that Mary wants to drink coffee.
- (e) I can see that when considering an implicit premise such as (1 d), the conclusion that Mary does not want to drink coffee at this time follows.
- (f) It is manifest to Mary that her utterance yields this conclusion if one accesses the (1 d) as implicit premise.
- (g) It is manifest that if Mary intends me to access this premise, she is justified in believing her utterance to follow the maxims.
- (h) Hence Mary conveyed (1 d) and (1 f) as implicatures.

³Frank and Goodman (2012), Goodman and Frank (2016), Goodman and Stuhlmüller (2013). For an introduction, see Degen (2023), Erk (2022) and Scontras, Tessler and Franke (n.d.).

⁴See also Grice's "working out schema" for implicatures in Appendix A.

Notice that this formulation differs considerably from the explanation of the very basic RSA model in Scontras, Tessler and Franke (Chapter 1 n.d.), which has the following layering: a “pragmatic listener” reasons about a “pragmatic speaker” who in turn reasons about a “literal listener”. However, it is unclear how a “literal listener” enters the Gricean schema of implicature identification in any useful way.⁵ Moreover, what Scontras, Tessler and Franke (n.d.) and Tessler, Tenenbaum and Goodman (2022) call a “pragmatic speaker” appears to be a misnomer and actually describes a “naive speaker”, i.e. a speaker who chooses an utterance considering how a hearer who attends only to the literal meaning of the utterance would understand (a “literal listener”).⁶ Such a conception is only ever appropriate for conversations in a language where literal meaning is all there is, and the only questions arising are how appropriate it is to apply meaning M to a given situation S. This is the case in the simple reference game: this language doesn’t have syntax, and it has only a handful of words which have extensional meanings. Even the most simple human conversations using natural language are radically different than this, and Grice was one of the first to actually point this out.

4.1.2 Communication and domain-specific regularities

Relevance theory fully adopts the general view of communication outlined in (15). However, it deviates from Grice’s account in two important respects: on the descriptive side, it claims that the explicit side of communication involves more inference than Grice imagined: in addition to pronoun reference and ambiguity resolution, the meaning of words need to be fine-tuned in context and linguistically unspecified variables need to be supplied before truth conditions can be evaluated. (For discussion, see Robyn Carston 1988; Robyn Carston 2002)

On the explanatory side, Sperber and Wilson (1995) seek to find the inference-guiding regularities of the conversational domain not in an implementation agnostic rational analysis (such as in Grice’s ‘working out scheme’ for implicatures, Grice (1989, pp. 30–31)) but in a psychologically adequate account of mental processes. As we have seen above, this account recognises relevance not only as a property of utterances, but of inputs to cognitive processes in general. It furthermore shows how a modest generalisation about the role of relevance in cognition (the cognitive principles of relevance in 7) has profound effects when the input to cognitive processes happens to be an overtly intentional (i.e. *ostensive*) act: in this case, inference follows a pattern (described in 9) that greatly reduces the processing steps needed for the audience to justifiably reach a conclusion about the speaker’s intended meaning.

Notice that the inference pattern in the comprehension heuristic in (9) specifies a rule to search through interpretive options (access candidates in order of accessibility, i.e. follow a path of least effort), defines a stopping criterion (check if this interpretation yields cognitive effects as expected) and provides a one-reason framework for decision making (attribute the first interpretation satisfying the stopping criterion to the speaker’s intention). In other words, the procedure appears to be an instance of what Gigerenzer (2004, p. 74) calls a *Take the Best* heuristic: a procedure that is simple,

⁵Therefore it seems to us that either (1) Scontra’s et al’s “vanilla RSA model” deviates from Gricean pragmatics at this point, or (2) their “vanilla RSA model” is applicable only to their very simple reference game and needs to be extended when considering more real life communicative behaviour.

⁶Such a “naive speaker” does not differ much from a “literal speaker”, but Tessler, Tenenbaum and Goodman (2022) already use the term “literal speaker” to a speaker who is producing conclusions to syllogisms literally.

exploits regularities of a specific domain (that ostensive stimuli convey expectations of their own optimal relevance), and allows successful inference with partial and incomplete knowledge.⁷ argue that the relevance-theory processing heuristic should be interpreted as such a “fast and frugal” cognitive heuristic in Gigerenzer’s sense and constitutes a mental module dedicated to the comprehension of ostensive stimuli.

Gigerenzer and colleagues appear to work from the fundamental assumption that probabilistic optimization models are inherently complex and therefore rather unlikely to be able to model mental processes directly. They provide the theoretical gold standard against which the performance of heuristic procedures can be measured, but do not provide insight into how cognitive processes work. If this fundamental assumption is justified, and if we accept Sperber and Wilson’s argument that the comprehension procedure they propose is a cognitive heuristic hard-wired into a comprehension module, then it follows that we should apply Bayesian modelling to relevance theory only at the higher level of modelling the conversation exchange. We could then model Mary and Peter’s conversational exchange as in (17):

- (17) Mary’s and Peter’s reasoning based on the assumption that listeners are hard-wired to apply the relevance-theory heuristic:

MARY: [“pragmatic speaker” reasoning about a “pragmatic listener”]

- (a) I want to make it manifest to Peter that I don’t want to drink coffee at 6 pm.
- (b) I also want to make it manifest to Peter that I do want something to drink and that I do want to be sociable at the party.
- (c) I need to choose the utterance that maximizes the probability that Peter’s first interpretive hypothesis that he tests following a path of least effort will make it more manifest to him that I have these two goals.

PETER: [“pragmatic listener” reasoning about a “pragmatic speaker”]

- (a) I expect Mary to confirm or reject that Mary wants to drink a cup of coffee at 6 pm.
- (b) I expect Mary to be sociable even when declining the offer.
- (c) The most easily accessible interpretation I get is the one in table 1 and it satisfies my expectations.

Notice how the assumption that the relevance-theoretic comprehension heuristic is hard-wired in the mental architecture of speakers and listeners radically simplifies the layout of the probabilistic modelling of conversation as opposed to (16).

The comprehension heuristic works on the assumption that different interpretive hypotheses vary in the cost incurred in processing them, and therefore in their order of accessibility. In probabilistic modelling, we can capture this effect by sampling from a set of interpretive hypotheses that are ordered in terms of accessibility. This order is arbitrarily determined by the experimenter to test the effect of various assumptions about the relative cost of constructing interpretive hypotheses. By operating with manually constructed hypothesis sets and accessibility orders we capture the idea

⁷Wilson and Sperber 2004, pp. 624–625.